



CHAPTER 1

Introduction



The only way to learn is by playing. The only way to win is by learning. And the only way to begin is by beginning.

— *Sam Reich* —





ARGUABLY THE GREATEST ADVANCEMENT in modern astronomy was the direct detection of gravitational waves – the rippling of space-time as a result of the motion of massive celestial bodies – on September 14, 2015. The Laser Interferometer Gravitational Wave Observatory (LIGO) collaboration detected the chirp of merging binary black holes, with masses $m_1 = 36^{+5}_{-4} M_\odot$ and $m_2 = 29^{+4}_{-4} M_\odot$ (Abbott et al., 2016). This detection sparked the so-called “gravitational wave era”, with detections in Operating Run 4 (O4) reading a cadence of hundreds per year.¹ These detections afford researchers probes of various physical phenomena. For instance, these merger events are some of the most energetic events in the Universe – GW150914 peaked at a gravitational-wave luminosity of $3.6 \times 10^{56} \text{ erg s}^{-1}$ (Abbott et al., 2016), compared with the typical energy release of a core-collapse supernovae (CCSN) of $\sim 10^{53} \text{ erg}$ (Hartmann, 1999) – and the merger of binary neutron stars are the predominant formation channels for most of the elements heavier than iron (see, among others, Eichler et al., 1989; Arcavi et al., 2017; Pian et al., 2017; Smartt et al., 2017).

In addition, the stochastic background of gravitational waves is a direct probe on early-Universe cosmology (Christensen, 2018), though the frequency band that these gravitational waves span is between 10^{-9} and 10^{14} Hz (Christensen, 1992, 2018), making detection of the stochastic background difficult. Indeed, it took eight years after the detection of gravitational waves for the stochastic background to be detected by the collaborative effort of North American Nanohertz Observatory for Gravitational Waves (NANOGrav) (Agazie et al., 2023), the European Pulsar Timing Array (EPTA) (Antoniadis et al., 2023), the Parkes Pulsar Timing Array (PPTA) (Reardon et al., 2023), and the Chinese Pulsar Timing Array (CPTA) (Xu et al., 2023).

Beyond the stochastic background, the vast majority of gravitational wave sources involve stars at some point in their lives – whether that be the merger of Compact Objects, the detonation of CCSN, or the rotation of an isolated massive source such as a Neutron Star (NS) or Black Hole (BH). In these cases stars are either the immediate sources of gravitational waves, or the progenitors of the objects that are the sources. Only in extreme cases, such as the merger of galaxies or the motion of supermassive black holes, are stars not directly involved in the generation of gravitational waves (and even then, this is debatable).

Ergo, any attempt at understanding gravitational waves must be founded in a strong theoretical understanding of stellar structure and evolution. Without theory, any observation is no more than a curiosity; it is with theoretical underpinning and models that insight can be gained. As most astrophysical sources of gravitational waves (i.e., not the stochastic background) are formed by the merger of two Compact Objects (COs) – black holes, white dwarfs, and neutron stars – one must understand their formation and dynamics, and thus the evolution of progenitor stars into these remnants, as well.

Furthermore, a sound understanding of the evolution of stars has impacts beyond gravitational wave astronomy. As the interiors of stars are complex nuclear furnaces, they are responsible for producing most of the elements in the Universe heavier than Hydrogen (Burbidge et al., 1957). In fact, of the elements lighter than Uranium, the only elements not substantially produced in stellar processes (such as through kilonovae or in supernovae) are technetium, promethium (which both lack stable isotopes),² boron, beryllium (which

¹From the start of O4 in May 2023 to the end of July 2025, approximately 219 unretracted significant (false alarm rate of $< 1 \text{ month}^{-1}$) detections, averaging to a cadence of $\sim 97 \text{ yr}^{-1}$.

²There are technetium stars (Merrill, 1952; Smith & Lambert, 1988; Uttenthaler & Lebzelter, 2010, among others), and

both are produced via cosmic ray fission), and some formation channels of lithium.

Starlight is also a critical requirement for probing the observable Universe. Photons are the means by which many astrophysical observations are made. Thus, a complete picture of the births, lives, and deaths of stars – and the compact remnants that they may leave after the supernova – is essential to understand spectra, nucleosynthetic yields, chemical abundances, and supernovae. These models also allow us to probe remnant mass distributions, characteristics of supernovae progenitors, and the binary interactions that define the majority of the stars in the Universe. This picture is incomplete without stellar structure and evolution models.

From the perspective of Universal structure and habitability, stars are also the focal point of planetary systems, and provide the theoretical right conditions for the formation of life. The protoplanetary discs that form around stars depend on the host star’s mass and luminosity. Without a complete picture of stellar evolution, one is unable to describe the likelihood of extraterrestrial life within the Universe.

1.1 | Observational astronomy

If the stars should appear one night in a thousand years, how would men believe and adore; and preserve for many generations the remembrance of the city of God which had been shown!

— “NATURE”, BY RALPH WALDO EMERSON

There are approximately 9000 stars visible in the night sky to the naked eye (with magnitudes brighter than 6.5), across both hemispheres (Hoffleit, 1964), which are displayed in Figure 1.1. The dark-adapted eye can see approximately as far as η Carinae,³ which is only approximately 2300 pc away.

2300 pc is an order of magnitude less than the size of Galaxy, which is approximately 26.8 kpc in diameter (Goodwin et al., 1998), which in turn pales in comparison to the Local Group at ~ 3 Mpc (Redd, 2018), and the observable Universe at 28.5 Gpc (Bars & Terning, 2010). There are, in addition, $\sim 1.7 \times 10^{22}$ stars in the observable Universe, assuming 1.7×10^{11} galaxies, with approximately 10^{11} stars in each (Marov, 2015). Thus, the birth, evolution, and death of stars is of fundamental importance to understanding the local structure and evolution of astrophysical objects on any scale.

Given the number of stars visible in the vast expanse of night, and their relatively stable positions in the sky, it is no surprise that they are a fixture in many religions and cultures around the world. Ancient Greece saw a difference in movement of some celestial objects, and called these $\pi\lambda\alpha\nu\eta\tau\eta\varsigma$, which morphed into *planētes* and eventually the English word “planets”. Babylonia knew of Halley’s comet and tracked observations of it

promethium has been observed in some atmospheres (Cowley et al., 2004), though the most stable isotopes produced in stellar nucleosynthesis have half-lives on the order of 10^5 yr (Tc; Uttenthaler & Lebzelter, 2010) or 10^1 yr (Pm; Cowley et al., 2004).

³This is a debated topic; we use the definition that a “star” includes binaries and that the star is in the catalogue of Hoffleit (1964), which rules out V762 Cas, the other commonly held “farthest star”.

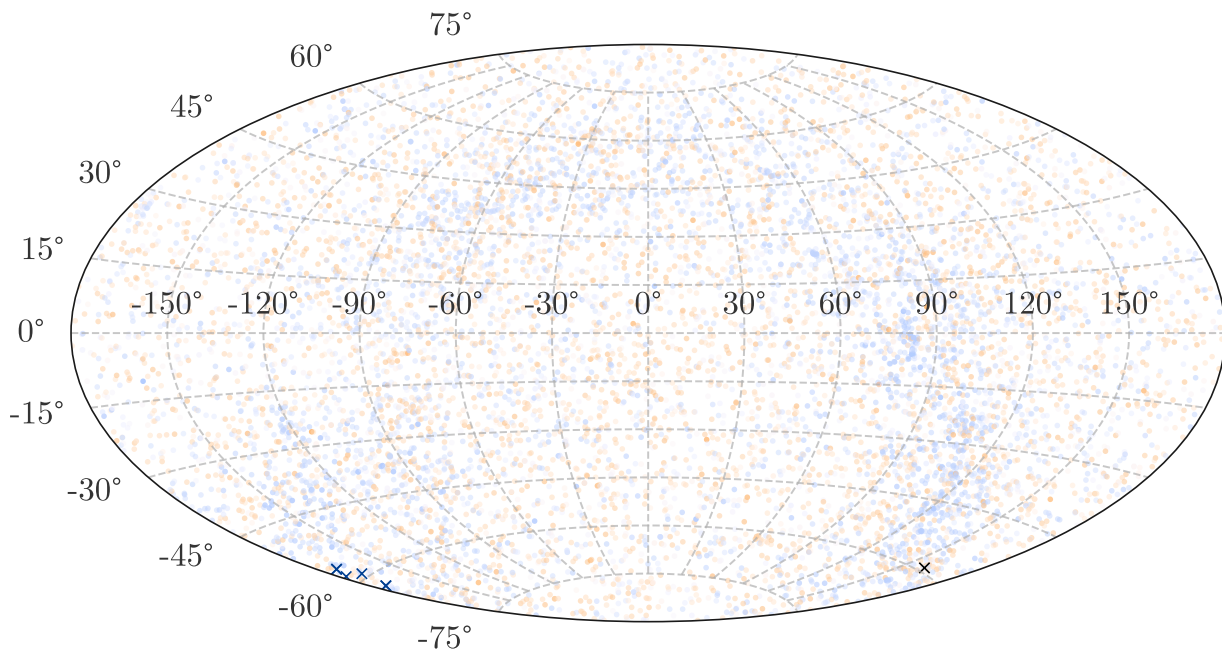


Figure 1.1: The locations of the ~ 9000 stars in the night sky visible to the naked eye, as reported in Hoffleit (1964). The colors are computed by converting the B-V color indices to temperatures using the fits in Ballesteros (2012), and then converted to RGB via the algorithm in this StackOverflow post: <https://stackoverflow.com/a/45497817>. The Southern Cross (Crucis) is shown fully opaque in blue towards the bottom left of the plot as a point of reference, and η Carinae is shown fully opaque in black in the bottom right.

as late as 87 BCE (Stephenson et al., 1985). Records of eclipses in Chinese culture exist as far back as 1600 BCE (Fang, 2015). The Nazca lines, geoglyphs carved into the Nazca desert in Peru at some point between 500 BCE and 500 CE, may be early astronomical calendars (Kosok & Reiche, 1949), though this theory was later placed into doubt (Hawkins, 1973). Knowledge of the movements of stars guided Polynesian wayfarers as they crossed the South Pacific (Buck, 1938; Hamacher, 2022), were embedded in the cultural myths of First Nations peoples in Australia (Noon & De Napoli, 2022), allowed them to forecast weather patterns (Hamacher et al., 2019), and were plausibly recorded into oral histories (Fuller & Hamacher, 2017). Knowledge of the lunar perigee was even written into *Othello* by Shakespeare:

It is the very error of the moon;
She comes more near the earth than she was wont,
And makes men mad.

– Shakespeare (ca. 1603)

Knowledge of oceanic tides existed in the time of Ancient Greece, and was described by Ptolemy, who wrote in *Tetrabiblos* that

ἀμπώτεις καὶ αἱ παλίσροιοι, καὶ αἱ τῶν πνευμάτων δὲ τροπαὶ μάλιστα περὶ τὰς τοιαύτας τῶν φωτῶν κεντρώσεις ἀποτελοῦνται πρὸς οὓς ἂν τῶν ἀνέμων ἐπὶ τὰ αὐτὰ τὸ πλάτος τῆς σελήνης τὰς προσνεύσεις ποιούμενον καταλαμβάνηται.	For the hour by hour intensifications and relaxations of the weather vary in response to such positions of the stars as these, in the same way that the ebb and flow of the tide respond to the phases of the moon, and the changes in the air-currents are brought about especially at such appearances of the luminaries at the angles, in the direction of those winds towards which the latitude of the moon is found to be inclining.
Ptolemy & Proclus (ca. 168)	Robbins (1940)

This knowledge flowed through to Dante, who wrote in the third part of the Divine Comedy, *Paradiso*:

E come 'l volger del ciel de la luna cuopre e discuopre i liti sanza posa, così fa di Fiorenza la Fortuna:	And, as the turning of the lunar sphere covers and endlessly uncovers the edges of the shore, thus does fortune deal with Florence.
Alighieri (ca. 1317)	Corbett & Webb (2016)

Despite this deep cultural connection to the stars, what we would now call Astronomy is a somewhat recent invention. Arguably conventional (and in some sense, “correct”) astronomy arose in the 16th century CE with Copernicus proposing the heliocentric model of the solar system, which was confirmed by Galileo in the 17th century CE (Carroll & Ostlie, 2017) – although Galileo’s refusal to recant his claims did result in his being convicted of heresy for 300 years. The 17th century brought about the establishment of orbital mechanics with the work of Kepler (based upon the observations of Tycho Brahe), and the discovery of gravity by Newton.

The early 20th century saw a veritable renaissance in both the observational and theoretical spheres. On the one hand, developments such as the theories of Special Relativity in 1905 (Einstein, 1905), General Relativity (GR) in 1917 (Einstein, 1917), and the predictions of gravitational waves (Poincaré, 1906; Einstein, 1916) led to theoretical revolutions. Around this time, Arthur Eddington began working on determining the internal structure and evolution of stars (Eddington, 1916, 1920), and Gamow (1928) put forth the idea of quantum tunneling allowing elements to overcomb the Coloumb barrier. On the other hand, the innovations of Payne (1925) in showing that stars are composed of mostly Hydrogen and Helium, Baade & Zwicky (1934) in distinguishing “super-novae” from “common novae”, Chandrasekhar (1935) in establishing the idea of the “Chandrasekhar mass” (the limiting mass of a white dwarf), Hoyle (1946) in developing the foundation of the

nuclear reaction of elements heavier than Hydrogen in stars, [Salpeter \(1955\)](#) in developing the Initial Mass Function (IMF), [Burbidge et al. \(1957\)](#) in truly developing the theory of stellar nucleosynthesis, [Heney et al. \(1955\)](#) and [Hayashi \(1961\)](#) in exploring the pre-main sequence evolution of stars were key to kickstarting entire branches of astronomy such as spectroscopy and computational astrophysics.

The pace of discovery of new and exotic astronomical phenomena has not slowed into the 21st century. Thorne-Żytkow Objects (TŻOs) – giant stars with NS cores ([Thorne & Żytkow, 1977](#)) – once only existed as theoretical curiosities, yet now have credible observational candidates in [Levesque et al. \(2014\)](#). The aforementioned gravitational waves were detected in 2015 ([LIGO Scientific Collaboration and Virgo Collaboration, 2016](#)), a century after they were demonstrated by [Einstein \(1916\)](#) to be a central prediction of GR. In the coming years, the Vera C. Rubin Observatory ([Ivezić et al., 2019](#)) and Laser Interferometer Space Antenna (LISA) are expected to vastly increase the number of short-period eclipsing double white dwarfs ([Burdge et al., 2020](#)), with the latter able to detect gravitational waves with frequencies between 0.1 mHz and 1 Hz ([Amaro-Seoane et al., 2012](#)), compared to the $\sim 10\text{ Hz} - 1000\text{ Hz}$ that LIGO-Virgo-KAGRA Collaboration (LVK Collaboration) can detect ([Schmitz, 2021](#)). The stochastic gravitational-wave background was detected by [Agazie et al. \(2023\)](#); [Antoniadis et al. \(2023\)](#); [Reardon et al. \(2023\)](#); [Xu et al. \(2023\)](#), via a search of pulsar-timing arrays.

Astronomy as a field has blossomed since the naked-eye era – entering the era of the telescope, the era of the observatory, and now the gravitational wave era. The next era of astronomy is the big data era. With observations reaching sub-day cadences, a large array of theoretical models, and perhaps the development of “on-demand” detailed models is required.

1.2 | Stellar Modelling

Modelling the evolution of a star is a nontrivial endeavour. Stars reach internal temperatures, pressures, and densities unsustainable on Earth. For instance, the core of a star is capable of nuclear fusion, which has not been sustainably recreated on Earth. Moreover, the density of Sirius B – a White Dwarf (WD) in the constellation of Canis Major – is $2.14 \times 10^{10}\text{ kg m}^{-3}$ ([Barstow et al., 2005](#)) and the mean density of the Sun is $\sim 1410\text{ kg m}^{-3}$ ([Eddy & Ise, 1979](#)). On Earth, the highest density measured in a laboratory is $\sim 27\,000\text{ kg m}^{-3}$, requiring pressures on the order of 2.3 TPa ([Fratanduono et al., 2020](#)), and observed for only nanoseconds.

It is nigh impossible to achieve and sustain these conditions to experimentally test any physical phenomena. Nevertheless, missions such as the Parker Solar Probe have made flyby missions into the atmosphere of the Sun, though broadening the sample size to two stars – using Proxima Centauri, the nearest star to Earth aside from the Sun – would require travelling at approximately $0.1c$ for 40 years to reach. If instead we were to send the Parker Solar Probe travelling at a constant velocity of its top velocity of $\sim 176\,471\text{ m s}^{-1}$, it would take approximately 7200 years to reach Proxima Centauri. On the other hand, meteorites do contain traces of presolar stardust that can be examined in a laboratory ([Clayton & Nittler, 2004](#)). Needless to say, physical

experimentation with a star is difficult.

1.2.1 | Stellar Modelling and Population Synthesis

Because of these factors, one often turns to computer simulations to model the interiors of stars. The question turns from physical experimentation to mathematical ones, and the idea of a *stellar model* becomes useful. A stellar model is a calculation performed such that the internal structure (the present state of physical quantities) and evolution (how these quantities change over time) are determined. Computing these parameters is nontrivial. As will be developed in Chapter 2, the simplest model of a star is a series of four coupled differential equations describing the internal pressure, temperature, luminosity, and mass distributions. However, including “extra” physics greatly increases the complexity of the problem; [Heney et al. \(1964\)](#) developed methods of solving the equations of structure and evolution that turn the problem into matrix inversion, at the cost of the central “Heney matrix” being thousands of elements in size.

The diversity and complexity of stellar profiles, interiors, and characteristics discussed earlier demands a careful and varied approach. One cannot abstract away much of the important physics. Although it is common to use approximations such as spherical symmetry to reduce the dimensionality of the problem, and to only evaluate the equations of stellar structure at a fixed number of locations in this one remaining dimension, chosen sufficiently intelligently, a complex computational effort must be undertaken.

This problem is then magnified when attempting to extrapolate out to describe an entire population of stars, such as from a burst of stellar formation activity, or in a shared environment such as a globular cluster. It is inefficient to, for example, model all approximately 10^{11} stars in the Milky Way ([Marov, 2015](#)) as independent entities. One must develop a way to take a library of models and determine the large-scale behaviour this way.

Broadly speaking, there are three approaches generating a population of stars, all of which have both merits and drawbacks. These three methods are:

1. Rapid Population Synthesis,
2. Detailed Population Synthesis, and
3. 3D Simulations.

Each of these approaches informs the other two, and all play important roles in stellar modelling. Although we focus on detailed methods, we will briefly outline all three approaches here.

Population Synthesis with Detailed Binary Models

A detailed population synthesis code uses detailed model sets, where the interior state of the stars is computed, and coupled to the orbital evolution equations. In this regime, observables such as mass transfer are explicitly calculated and solved for from the exact equations of state, structure, and evolution of the binary. By doing

this, one follows the binary evolution in a reasonable amount of detail, obfuscating fine detail and (generally) asymmetry in the star in favour of recovering the larger-scale structure.

An example of such a code is the BPASS code (Eldridge et al., 2017), which uses a custom version of the Cambridge STARS code for the stellar modelling and a bespoke proprietary code named TUI for the population synthesis. The key approach in the present version of BPASS is to model the primary star in detail until it undergoes a supernova, whilst approximating the evolution of the secondary. After the primary supernova, the secondary is then evolved in detail with the primary treated as a point mass. After the secondary supernova, the parameters of the remnants – if any – can thus be fed into a postprocessing code to have their orbital equations solved for further.

Rapid Binary Population Synthesis

Rapid binary population synthesis models are instead either compilations of single-star evolutionary tracks that have been solved for using a code such as MESA (Paxton et al., 2011, 2013, 2015, 2018, 2019; Jermyn et al., 2023), fit to using approximations such as those of Hurley et al. (2000) or Hurley et al. (2002), or interpolated between. These models are then coupled to the orbital evolution *a posteriori* using a binary evolution framework, effectively approximating the computationally expensive part of binary evolution models to varying degrees of detail. This allows them to generate binary models on-the-fly, and optimise for this interaction, at the potential cost of missing more precise facets of binary evolution.

3D Simulations

3D simulations are the most detailed method of stellar modelling. The base principle is that the star is evaluated as a network of particles in a continuous media, and the evolution of the star is followed in theoretically atomic detail. For example, PHANTOM (Price et al., 2018) is a Smoothed-particle hydrodynamics (SPH) code which essentially models the star as a fluid, where the Lagrangian hydrodynamical equations are solved for in two-dimensional cells (termed “particles”). Although these present the most detailed models, they are also the most time-intensive to solve. For a SPH code, the famous Navier-Stokes equation is invoked, which has no analytical solution but is computed in excruciating detail. As a result, most SPH models take weeks to run and simulate a star for kiloseconds of its lifetime, rather than for the millions (if not billions) of years detailed models are able to compute. 3D simulations are thus best placed to solve what are fundamentally *3D challenges*; these are processes such as thermohaline mixing and mass transfer. In a 1D code such as BPASS, these important physical processes must be approximated in some fashion. Full population synthesis using 3D simulations is also thus unlikely to be tenable without significant advances in computational techniques.

Though it may seem that they serve the same niche and are therefore competing to be the “better” method of stellar modelling, detailed 1D models and 3D models exist harmoniously with each other. The advancements in 3D modelling inform the evolution of 1D models in a process known as “3-to-1” (Rizzuti et al., 2023, 2024), and the computational flexibility of 1D models offers insights into where 3D models wish to target to evolve, in a lock-step advancement method.

1.3 | This work

We now describe the placement of this work within the field, and outline the key aims this thesis sets out to answer. This thesis can be summarised as an endeavour to model the interior physics of interacting binary stars over certain near-extreme phenomena. We also aim to examine the post-supernova kinematics of binary systems. Thus, the area of interest of this thesis is effectively the entire lifetime of a binary system post-birth, with the critical exceptions of the moment of supernovae, and the interior physics of compact objects.

1.3.1 | Research themes

There are essentially two major themes guiding this research.

- I. Can we study contact binary systems, and refine our treatment of Roche Lobe overflow (RLOF) to overcome numerical issues inherent in these models?
- II. Can we develop a testing regime for the natal supernova kick, which relies on more observations than just isolated pulsar velocities?

1.3.2 | Thesis outline

Six impossible things. Count them, Alice.

— ALICE IN WONDERLAND

This thesis is, broadly, a tour of the structure and evolution of binary stars from Zero-Age Main Sequence (ZAMS) to their eventual merger, and has been structured to approximately follow this life cycle. The thesis is structured as follows:

1. In Chapter 2, we provide the necessary theoretical background for future chapters, critically examine the pieces of software and numerical methods employed in this work, and discuss the logistics of running this single-thread grid-based code as a multiprocessed code on a distributed cluster.
2. In Chapter 3, we explore the regime in which mass transfer is assumed to become helium-rich, and the effects on the future evolution of these binaries. (Theme I)
3. In Chapter 4, we examine the kinematics of the post-supernova inspiral and the mechanics of the natal supernova kick. (Theme II)
4. In Chapter 5, we present the results of our stellar evolution codes in the interesting case when both stars almost (or in some instances, do) touch. (Theme I)
5. In Chapter 6, we review the thesis and conclude. (Themes I, II)
6. In Appendix A, we present additional derivations for key theoretical expressions from Chapter 2.

7. In Appendix B, we list the codes employed and their versions, as well as the default parameter choices we make for our stellar evolution models (Theme I).
8. In Appendix C, we present period-eccentricity distributions for the LVK Collaboration O3 contour (Theme II).
9. In Appendix D, we present the bespoke contact binary models that we discuss in Chapter 5 (Theme I).
10. In Appendix E, we enumerate the predicted remnants produced by the overcontact binaries in Chapter 5.
11. In Appendix F, we provide the KAITIAKI code necessary to generate the models for certain stellar evolution plots throughout the thesis. (Theme I)

We now move into the more detailed background, and outlay the theory that underpins all of our models.